

PHYSICS

# Notable Faculty & Student Achievements

## March 2026

Assistant Professor of Physics **Cody Crosby** co-authored an open-access article, titled "Digital light processing 3D printing enables versatile fabrication of human engineered heart tissues," that was recently published in *Cell Biomaterials*. Briefly, the authors demonstrated that engineered heart tissue (EHT) molds from inexpensive polymer resins could be deployed to fabricate heart tissue with enhanced cytoskeletal and metabolic maturity. The work can be accessed [here](#).

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## November 2025

Assistant Professor of Physics **Cody Crosby** co-authored "Dynamic Regulation of Granular Hydrogels Through Guest-Host Interactions to Spatiotemporally Guide Cellular Migration" in the journal *Advanced Science* (IF = 14.1), along with several postdoctoral scholars and graduate students in the Burdick Lab at the University of Colorado–Boulder. The manuscript represents work undertaken during his pre-tenure sabbatical and demonstrates that printing granular hydrogels in particulate suspension paths can create a dynamic environment that can regulate cell migration. The work can be found, open access, at [this link](#).

## October 2025

Professor of Physics **Steven Alexander** and four of his research students — **Heidi Jackson** '26, **Casandra Nunez** '26, **Anshika Katrodia** '26, and **Brooke Ramey** '27 — attended the Fall 2025 meeting of the Texas Section of the American Physical Society, held October 10-11 in Fort Worth. Dr. Alexander presented a poster titled "Using Molecular Annealing to Find Molecules with Specific Absorbance Peaks" and Brooke Ramey presented a poster titled "Verifying the Experimental Hyperpolarizabilities of Organic Molecules Using Density Functional Theory." Heidi Jackson, Casandra Nunez, and Anshika Katrodia presented a poster titled "Analyzing Emission Spectra to Determine the Mass of a Black Hole." All of the posters generated considerable interest and some insightful discussions.

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Assistant Professor of Physics **Cody Crosby** and three of his SURF research students — **Sabina Martinez Carreon** '27, **Bryan Guzman-Sanchez** '26, and **Rishi Rajesh** '27 — attended the 2025 Biomedical Engineering Society (BMES) Annual Meeting, held October 8-11 in San Diego. Dr. Crosby moderated two sessions at the international conference, a bioprinting session and a rapid-fire talk session for undergraduates in biomaterials. Sabina presented the group's poster, titled "Post-Print Reinforcement of GelMA/HAMA Scaffolds via PEGDA Infiltration," at the main conference and at the separate LatinX in BME Symposium. Both posters were well-attended by other conference-goers and generated some insightful and helpful discussions. **Amanda Mejia** '27 and **Gabriela Nicole Hislop Gomez** '26 also attended and participated in the conference.

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Associate Professor of Computer Science **Jacob Schrum** recently attended the Genetic and Evolutionary Computation Conference in Málaga, Spain to present his peer-reviewed paper, "A Quality Diversity Approach to Evolving Model Rockets," co-authored with Assistant Professor of Physics **Cody Crosby**. The article, freely accessible here from the ACM Digital library, describes the use of AI methods inspired by evolution to create designs for model rockets. An informative video about this joint research is also available here.

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## May 2025

Assistant Professor of Physics **Cody Crosby** first authored "Fabrication of Microgel-Reinforced Hydrogels via Vat Photopolymerization" in the journal *ACS Macro Letters* with Abhishek Dhand and the Burdick Lab at the University of Colorado Boulder. The manuscript is the culmination of his pre-tenure sabbatical and demonstrates for the first time that micron-sized hydrogels can be incorporated at high densities in complex resin-printed geometries. The work can be found here.

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## April 2025

author work is currently under review.

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Physics majors **Joseph (Joe) Dorsey** '26 and **Amanda Mejia** '27 attended the Capital of Texas Undergraduate Research Conference (CTURC) hosted at the University of Texas at Austin. Joe presented platform presentations on his and Amanda's research conducted in the bioprinting laboratory of Assistant Professor of Physics **Cody Crosby**.

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## March 2025

Associate Professor of Computer Science **Jacob Schrum** and Assistant Professor of Physics **Cody Crosby** had their paper, "A Quality Diversity Approach to Evolving Model Rockets," accepted for publication in the *Proceedings of the Genetic and Evolutionary Computation Conference*. For this paper, evolutionary computation algorithms studied by Dr. Schrum were applied to the problem of designing model rockets, like those built by Dr. Crosby's students in his Introduction to Engineering class. The designs evolved by Dr. Schrum's program were simulated in a computer, and the most promising designs were built by physics and computer science double-major **Kade Townsend** '25, before being launched and evaluated in the real world.

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November 2024

**Ashlyn Cadena '27, Olive Forrest '26, Andres Garza '27, Amanda Mejia '27, Olivia Johnson '26,** and Professor of Chemistry **Emily Niemeyer** attended the 2024 National Science Foundation (NSF) S-STEM Scholars and Principal Investigator Meeting in Chicago, IL. In addition to interacting with other S-STEM Scholars from around the country and attending workshops and professional development sessions, three Scholars presented their research in conference sessions. Garza's poster, titled "Laser induced synthesis of cobalt oxide oxygen evolution catalysts," described work he completed during a summer Research Experience for Undergraduates (REU) program at Texas State University. Mejia presented a poster, "Granular hydrogel synthesis for comparable in vitro model of brain parenchyma tissue," which is based on her ongoing research with Assistant Professor of Physics **Cody Crosby**. Cadena's poster, "Perimeter Minimizing Rectangles in  $R^2$  with density  $M|x| + N|y|$ ," described her collaborative research with other Southwestern math students, Associate Professor of Mathematics **John Ross**, and Adjunct Professor of Mathematics **Lauren Ross**.

Physics majors **Joseph Dorsey** '25 and **Amanda Mejia** '26 presented their extrusion bioprinting poster at the 2024 Regional Symposium (Southwest) hosted at UT Austin. Simultaneously, Assistant Professor of Physics **Cody Crosby** presented his lab's pedagogical work on polymer modification at the 2024 Regional Symposium (Western) hosted at the University of Colorado Denver (Anschutz campus).

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## October 2024

**Dante Medina** '26 presented a poster titled "Using Molecules to Store Energy" at the Southwest Regional Meeting of the American Chemical Society in Waco, TX. This presentation resulted from research done with Professor of Physics **Steve Alexander**.

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**Dominic Mashak** '26 presented a poster titled "Finding Molecules with Large Hyperpolarizabilities" at the Southwest Regional Meeting of the American Chemical Society in Waco, TX. This presentation resulted from research done with Professor of Physics **Steve Alexander** during the summer of 2024.

## July 2024

Assistant Professor of Physics **Cody Crosby** co-authored “Endometrial decidualization status modulates endometrial microvascular complexity and trophoblast outgrowth in gelatin methacryloyl hydrogels” in the journal *npj Women’s Health* with Dr. Samantha Zambuto and the Harley Lab at the University of Illinois at Urbana-Champaign. The work was published open-access and can be downloaded [here](#).

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## April 2024

**Gabriela Nicole Hislop Gomez** '26, **Angel Rodriguez** '24, and **Sabrina Woodward** '25 gave platform presentations at the Capitol of Texas Undergraduate Research Conference (CTURC) hosted at UT Austin. Their presentations were the result of work conducted in the bioprinting laboratory of Assistant Professor of Physics **Cody Crosby**.

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Assistant Professor of Physics **Cody Crosby** co-authored "Fabrication and Characterization of Quad-Component Bioinspired Hydrogels to Model Elevated Fibrin Levels in Central Nervous Tissue Scaffolds" in the journal Gels with Dr. Dany Munoz-Pinto and his undergraduate laboratory at Trinity University. The work was published open-access and can be downloaded [here](#).

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**Gabriela Nicole Hislop Gomez '26**, **Noor Nazeer '24**, and **Angel Rodriguez '24** presented their research talks at the Texas Academy of Science (TAS) 2024 Annual Meeting hosted at UT Permian Basin in Odessa. Noor and Angel were awarded first place in the chemistry/biochemistry and physics/engineering sections, respectively. Their presentations were the result of work conducted in the bioprinting laboratory of Assistant Professor of Physics **Cody Crosby**.

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## February 2024

Assistant Professor of Physics **Cody Crosby** published a review titled "Open-source extrusion 3D bioprinters: Trends and recommendations" in the journal Bioprinting. The publication evaluates the latest syringe extruders that have been designed to extrude bio-inks and offers concrete recommendations to ensure that this technology remains inexpensive and open-source. The process involved building and testing the five most popular designs available in the literature, as seen in Figure 2. The paper can be accessed for free for 50 days [here](#).



Assistant Professor of Physics **Cody Crosby**, alongside three outstanding Southwestern alumni and current students (**Domenic Cordova** '23, **Angel Rodriguez** '24, and **Nina Woodward** '24), published an article in HardwareX that demonstrated the development of an open-source bioprinting extruder for Ender-series 3D printers. <https://doi.org/10.1016/j.ohx.2024.e00510>

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## November 2023

Five students presented four talks at the Texas Undergraduate Mathematics Conference (TUMC), and four more students attended. The conference was held at Stephen F. Austin State University in Nacogdoches, Texas, on October 27-28, 2023. **Adrianna Flores-Vivas** '24 presented "Wolf Reintegration in Yellowstone National Park." **Ashley Odell** '24 and **Madison Williams** '24 presented "Does Money Really Buy Happiness?". **Blue Goodson** '24 presented "The Mathematical Artistry of Portrait Making." **Johanna Campbell** '24 presented "To the Heart of the Milky Way." The speakers gave preliminary presentations of their mathematics capstone projects under the supervision of Associate Professor of Mathematics **Therese Shelton**. Associate Professor of Physics **Mark Bottorf** co-supervised Campbell's. Shelton also moderated two sessions of student talks. **Zoe Kincaid** '24 attended, and first-year S-STEM/EQUIP students **Amanda Mejia** '27, **Juliana Elizondo** '27, and **Alyanna Martinez** '27 also attended. The TUMC is partially supported by the National Science Foundation grant DMS-2226539. The Atkin Junior Professorship in Mathematics for Assistant Professor of Mathematics **Noelle Sawyer** provided funding.

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Biomaterial Inks to Determine Significant Printing Parameters,” earned complimentary reviews from the graduate student judging panel.

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## October 2023

Assistant Professor of Physics **Cody Crosby** and four of his FSP research students— Nicole Hislop '26, Noor Nazeer '25, Angel Rodriguez '24, and Nina Woodward '25 — attended the 2023 Biomedical Engineering Society (BMES) Annual Meeting, held October 11-14 in Seattle, Washington. They presented two posters focused on the development of a novel biomaterial ink for simulating brain parenchyma tissue and the fabrication of a syringe extruder for Ender series 3D printers (the “Enderstruder”). Both posters were well-attended and generated some fascinating discussions.

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## May 2023

Kristie Cheng '23 and Angel Rodriguez '24 presented their work with Assistant Professor of Physics **Cody Crosby** on March 3-4 at the 2023 Texas Academy of Sciences (TAS) Annual Meeting hosted at San Angelo State in San Angelo, Texas. Cheng presented a talk entitled “Printability of biomaterial inks to mimic the blood-brain barrier,” and Rodriguez presented a talk entitled

## March 2023

Domenic Cordova '23 and Nina Woodward '25 presented their work with Assistant Professor of Physics **Cody Crosby** on March 4 at the Capitol of Texas Undergraduate Research Conference (CTURC), hosted on UT Austin's campus. Cordova presented a poster entitled "A Low-Cost Affordable Bioprinter Attachment for an Ender 3 3D Printer," and Woodward presented a poster entitled "An ImageJ Script to Analyze the Printability of Bioprinted Lattices." Both earned complimentary reviews from the judging panel.

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## December 2022

A wonderful interdisciplinary group from Southwestern participated in the 17th Annual Texas Undergraduate Mathematics Conference (TUMC), held this year at the University of Texas at Austin on October 29. Carson Vogel '23 presented "Modeling Heat Transfer." This project is a continuation of the 2021 and 2022 SCOPE projects under the supervision of Professor of Physics **Steven Alexander** and Associate Professor of Physics **Mark Bottorf**. This work is part of ongoing efforts for the eventual development of a solar energy storage cell; a problem brought to Southwestern by Coordinator of Science Facilities and Equipment **Oscar Lee Fellows**. Melanie Richey '23 presented "Rats on the Run: Modeling of Hippocampal Cell Activity Using Plasticity." Her project is a continuation of a 2022 Research Experience for Undergraduates at Southern

attended the conference. Marr chaired a student presentation session. Shelton supervised Vogel's and Richey's current work, preliminary results for their mathematics capstone projects. Jillian Reese '23 and Emma Lewis '23 joined with their counterparts from the University of North Texas-Denton in research with Shelton and Dr. Joe Iaia, funded through the Council for Undergraduate Research in Mathematics. Other students also attended: Oliver Johnson '23, Jess Kazmir '23, Lauren Calzado '23, Rowan Via '23, Kathryn Altman '24, and Aidan Bujanda-Moore '23. Majors and minors among our student attendees included Mathematics, Computational Mathematics, Physics, Economics, Education, Spanish, and Political Science.

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## November 2022

Assistant Professor of Physics **Cody Crosby** and two of his FSP research students— Domenic Cordova '23 and Noah Pyles '23 — attended the Great Plains Biomaterials Day in Lawrence, Kansas, on November 4, 2022. They presented two rapid-fire talks focused on developing a novel biomaterial ink for the blood-brain barrier and the fabrication of a low-cost Ender-3 bioprinter. Only 5 of the 35 submissions were selected for rapid-fire talks at the conference.

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## October 2022

continuation of the 2021 and 2022 SCOPE projects under the supervision of Professor of Physics **Steven Alexander** and Associate Professor of Physics **Mark Bottorf**. This work is part of ongoing efforts for the eventual development of a solar energy storage cell; a problem brought to Southwestern by Coordinator of Science Facilities and Equipment **Oscar Lee Fellows**. Melanie Richey '23 presented "Rats on the Run: Modeling of Hippocampal Cell Activity Using Plasticity." Her project is a continuation of a 2022 Research Experience for Undergraduates at Southern Methodist University under the supervision of Dr. Katie Hedrick in collaboration with Dr. Brad E. Pfeiffer, Department of Neuroscience, UT Southwestern Medical Center. Professor of Mathematics **Alison Marr** and Associate Professor of Mathematics **Therese Shelton** also attended the conference. Marr chaired a student presentation session. Shelton supervised Vogel's and Richey's current work, preliminary results for their mathematics capstone projects. Jillian Reese '23 and Emma Lewis '23 joined with their counterparts from the University of North Texas-Denton in research with Shelton and Dr. Joe Iaia, funded through the Council for Undergraduate Research in Mathematics. Other students also attended: Oliver Johnson '23, Jess Kazmir '23, Lauren Calzado '23, Rowan Via '23, Kathryn Altman '24, and Aidan Bujanda-Moore '23. Majors and minors among our student attendees included Mathematics, Computational Mathematics, Physics, Economics, Education, Spanish, and Political Science.

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Assistant Professor of Physics **Cody Crosby** and four of his FSP research students— Kristie Cheng '23, Domenic Cordova '23, Noah Pyles '23, and Angel Rodriguez '24— attended the 2022 Biomedical Engineering Society (BMES) Annual Meeting, held October 12-15 in San Antonio, Texas. They presented two posters focused on the development of a novel biomaterial ink for the blood-brain barrier and the fabrication of a low-cost Ender-3 bioprinter. Both posters were well-attended by other conference-goers and generated some insightful and helpful discussions.

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**Jonathan Smart** '23 presented his work with Assistant Professor of Physics **Cody Crosby** titled "Quantifying the Printability of Biomaterial Inks" on April 23 at the Heart of Texas Undergraduate Research Conference sponsored by Baylor Undergraduate Research in Science & Technology.

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## November 2021

Assistant Professor of Physics **Cody Crosby** gave a poster presentation titled "Phototunable Interpenetrating Polymer Network Hydrogels Stimulate iPSC-EP Vasculogenesis" at the Tissue Engineering and Regenerative Medicine International Society 6th World Congress, held virtually November 15–19.

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## September 2021

The lab of Assistant Professor of Physics **Cody Crosby** was recently featured in a KXAN News story, which was picked up by NBC affiliates in Denver, Colorado, and Albuquerque, New Mexico.

## April 2021

Business and biology major **Andrew Vergote** '21 gave a talk titled "Novel Bioinks: The Gateway to Bioprinting Complex Biological Tissues" at the 2021 BBB South Central Virtual Regional Convention. The talk resulted from research that Vergote completed with Assistant Professor of Physics **Cody Crosby** in spring 2021. They plan to continue this research as a funded Faculty–Student Project this summer.

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Assistant Professor of Physics **Cody Crosby** presented "Phototunable Interpenetrating Polymer Network Hydrogels Stimulate iPSC-EP Vasculogenesis" during two Rapid-Fire Sessions, which feature 10-minute talks, at the Rock Stars of Regenerative Engineering Conference and the Annual Meeting of the Society for Biomaterials, which were held virtually April 19–23, 2021.

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Physics and mathematics major **Gerardo Gonzalez** '22 gave a talk at the 2021 spring meeting of the American Physical Society (Texas Section) titled "Transition Probabilities for a Relativistic One-Electron Atom." Most of the research presented during this talk resulted from a 2019 SCOPE project that Gonzalez completed with Professor of Physics **Steven Alexander**. They are currently working on a paper that will describe their calculations.

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## January 2021

Assistant Professor of Physics **Cody Crosby** had his article " Phototunable Interpenetrating Polymer Network Hydrogels to Stimulate the Vasculogenesis of Stem Cell–Derived Endothelial Progenitors " accepted for publication in *Acta Biomaterialia* (with an impact factor of 7.242). The preprint can be accessed at [bioRxiv](#) .

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## April 2020

Double major in German and physics **Claire Harding** '20 earned a prestigious Fulbright Award. The nationally competitive Fulbright English Teaching Assistantship will take Harding to Germany, where she will integrate hands-on science learning with lessons in American cultures and the English language. The Fulbright English Teaching Assistant (ETA) Program places recent college graduates and young professionals as English teaching assistants in secondary schools or universities overseas, improving foreign students' English language abilities and knowledge of the U.S. while increasing the U.S. student's own language proficiency and knowledge of the host country. ETAs also pursue individual study/research plans in addition to their teaching responsibilities. Harding's teacher and mentor, Associate Professor of German **Erika Berroth**, deeply appreciates Harding's contributions to the German program as a tutor and member of the leadership team of the German Club.

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